

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 19-20-21(Ch. 16 + GF, Ch.4, Q8)
Expectations, Consumption and Investment

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model
 1. Medium and short run equilibrium
 2. Demand and supply shock

Extensions

4. **Expectations, financial markets, and economic policies**
 1. Financial markets and expectations
 2. **Consumption, investment, and expectations**
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies

EXPECTATIONS AND REAL MARKETS

Expectations about future macro context (GDP, MP, FP) play a role on consumption and investment decisions

=> influence current demand, production and income

How is aggregate consumption determined?

Two approaches

1. **Traditional Keynesian theory:** current consumption depends on current disposable income

$$C_t = f(Y_t - T_t)$$

2. **Intertemporal micro founded theory:** consumption depends on total wealth in the life cycle (Modigliani) or permanent income (Friedman)

$$C_t = f(W_t^T)$$

All the resources in the life cycle:

- Human wealth
- Financial wealth
- Housing wealth

In each period, a fraction of total wealth is spent in order to maintain a relatively constant level of consumption over time

Consumers are very foresighted since at time t

- *Know* their current disposable labor income and current short term interest rate

$$Y_{Dt} = Y_t - T_t \quad r_t = i_t - \pi_t$$

- *Have expectations* about their future disposable income and future short term interest rate

$$Y_{Dt+1}^e = Y_{t+1}^e - T_{t+1}^e \quad r_{t+1}^e = i_{t+1}^e - \pi_{t+1}^e$$

⇒ Total lifetime income = expected present discounted value of current and future (disposable) after tax income

$$LW_t = (Y_t - T_t) + \frac{1}{1 + r_t} (Y_{t+1}^e - T_{t+1}^e)$$

Consumption will depend on total wealth

$$W_t^T = \underbrace{LW_t}_{\substack{\text{Human (labor) wealth:} \\ \text{expected PDV of after tax} \\ \text{income over working life}}} + \underbrace{FW_t + HW_t}_{\substack{\text{Non human wealth}}} +$$

Financial wealth:
net value of asset holdings
(assets – liabilities)

Housing wealth:
value of houses – mortgage

For the single worker it depends on human capital that influences productivity and wages and at the aggregate level it $\uparrow$ if

- Current/expected income $\uparrow$
- Current/expected taxes $\downarrow$
- Current/expected interest rate $\downarrow$

Macro context

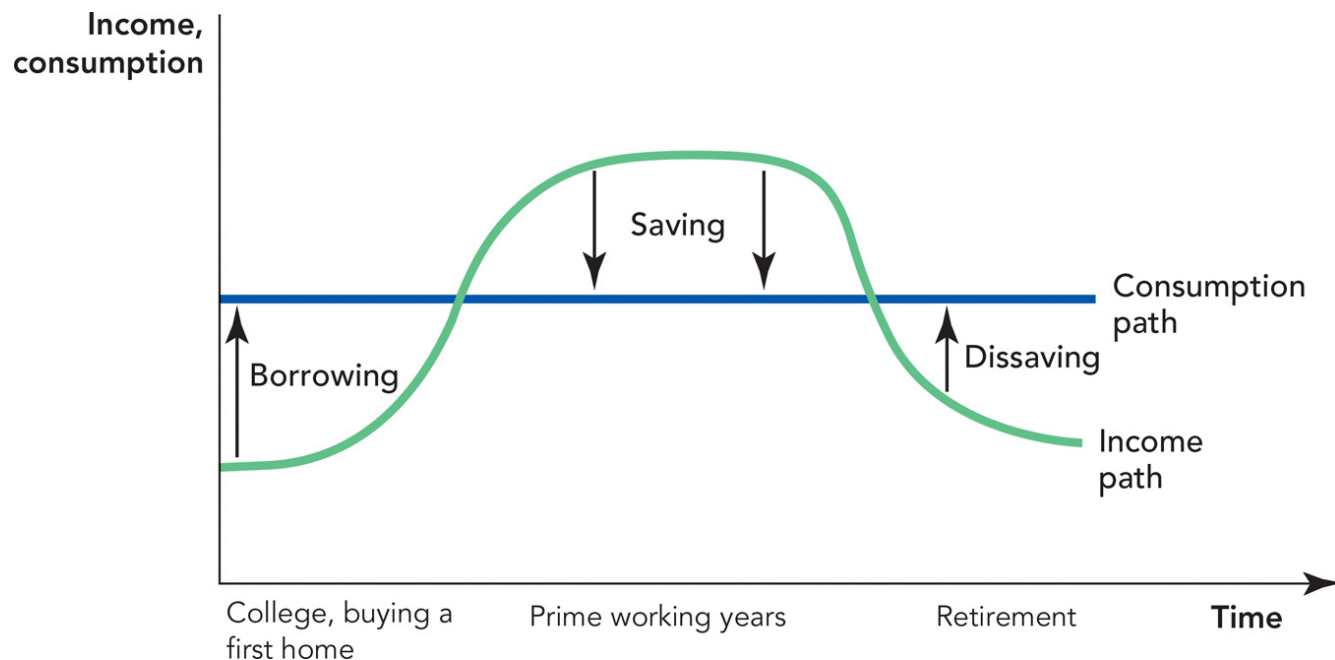
The foresighted consumer spend in each period a fraction of the total lifetime wealth

$$C = f(W_t^T) +$$

How much is consumed in each period?

- Desired consumption more or less stable over time
- Income fluctuates over time

- ⇒ Agents compare their desires with their available resources
- ⇒ make consumption smoothing: the fraction of total wealth consumed stays relatively stable over time, thanks to saving and borrowing (they allow to transfer resources over time)



MICRO-FOUNDATION

INTERTEMPORAL CONSUMPTION CHOICES

The optimal intertemporal consumption path maximizes intertemporal utility

Assumptions

- Two period: present (t) and future (t+1)
- Utility function: increasing and concave in the level of consumption in the two periods

$$U = f(C_t, C_{t+1})$$

- Free access to the credit market:

r = real interest rate

cost of borrowing/return to saving in real terms (units of goods)

- Only human wealth (no financial nor housing)

What is the optimal intertemporal consumption choice?

What determines whether the individual is a borrower, a lender or consumes exactly disposable income?

From the micro approach we know that it depends on

- Preferences
- Constraints: interest rate and available resources (current and expected)



Intertemporal budget constraint:

obtained from the budget constraint in each period

It represents the trade-off consumers face between consumption today and consumption in the future: all possible affordable combinations of consumption across different time periods, given their income in each period and the prevailing interest rate

- Present BC, t

$$Y_t - T_t = C_t + S_t$$

$$S_t = Y_t - T_t - C_t$$

If $S > 0$: saver

$S < 0$: borrower

- Future BC, $t + 1$

$$C_{t+1} + S_{t+1} = (Y_{t+1}^e - T_{t+1}^e) + (1 + r_t)S_t$$

Being the last period, it makes no sense to save ($S_{t+1} = 0$)

⇒ Substituting current saving in the future BC:

$$C_{t+1} = (Y_{t+1}^e - T_{t+1}^e) + (1 + r_t)(Y_t - T_t - C_t)$$

$$(Y_t - T_t) + \frac{1}{1 + r_t}(Y_{t+1}^e - T_{t+1}^e) = C_t + \frac{1}{1 + r_t}C_{t+1}$$

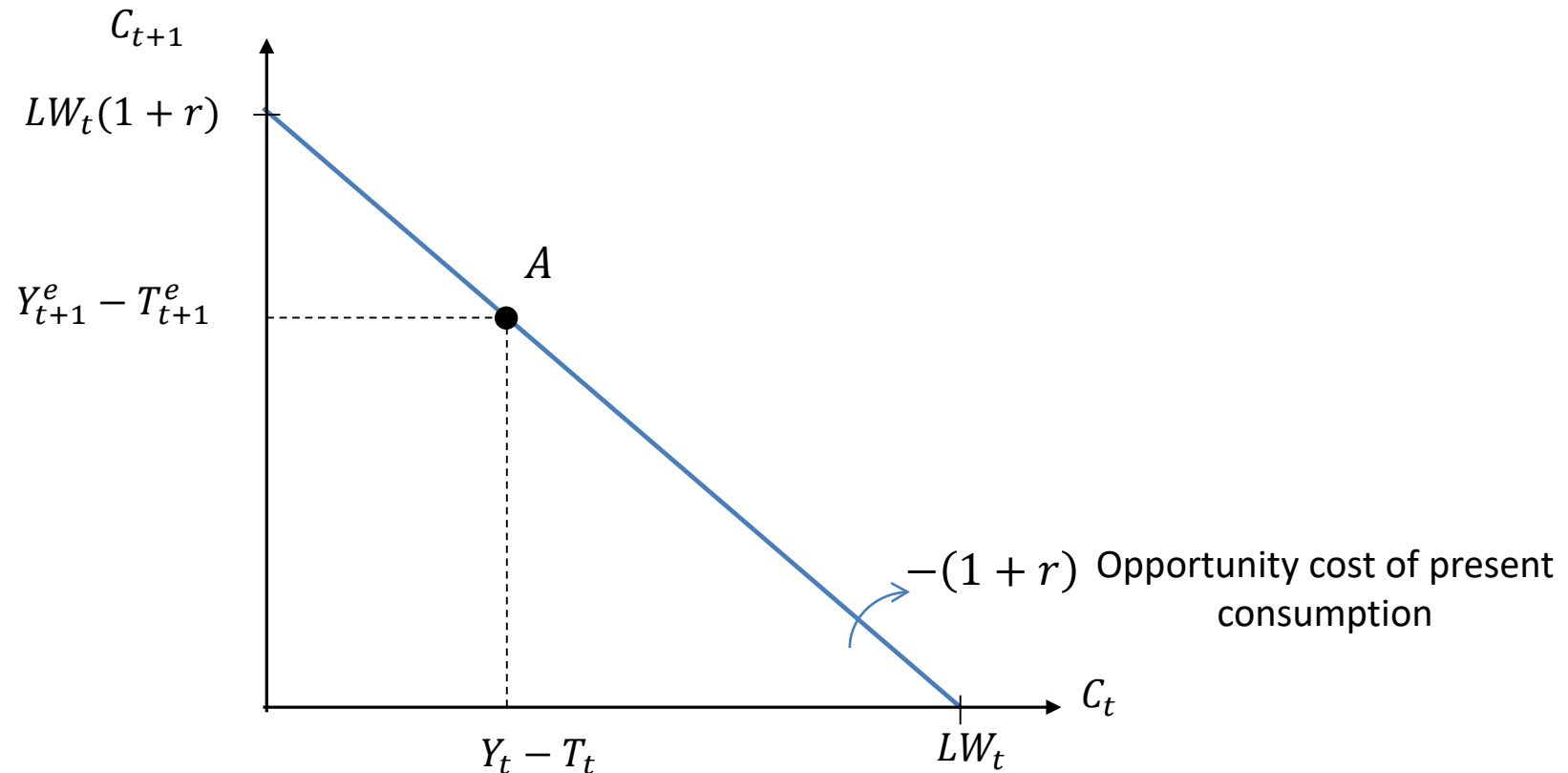
LW_t = PDV of lifetime
income or
Permanent income

Present discounted value
of consumption over the
two periods

$$\text{IBC: } C_t + \frac{C_{t+1}}{1+r} = LW_t \quad \Rightarrow \quad C_{t+1} = LW_t(1+r) - (1+r)C_t$$

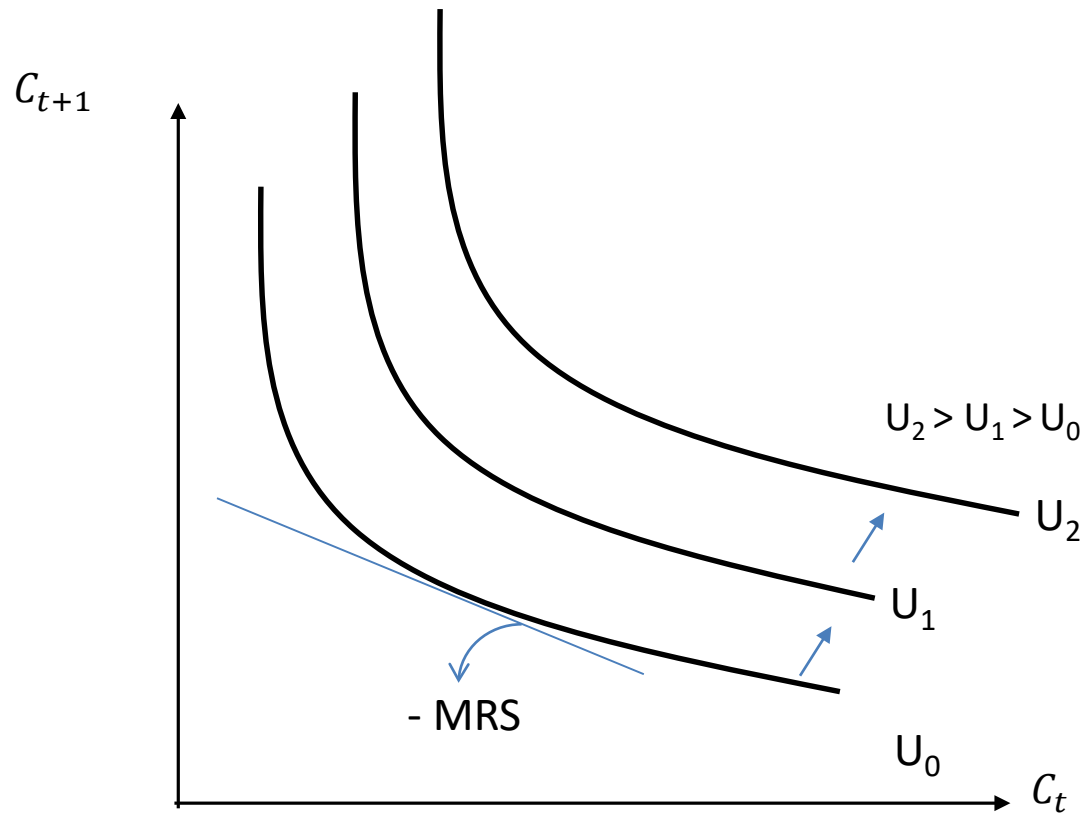
All possible intertemporal consumption paths represented in the (C_t, C_{t+1}) plane as a line with

- Negative slope
- Passing from the point of the initial endowments in the two periods



Preferences: utility function

- $U \uparrow$ along curves farther from the origin
- Negative slope (MRS): trade-off btw current and future consumption while keeping utility constant
- Convex: diminishing MRS



Optimal intertemporal consumption path maximizes utility

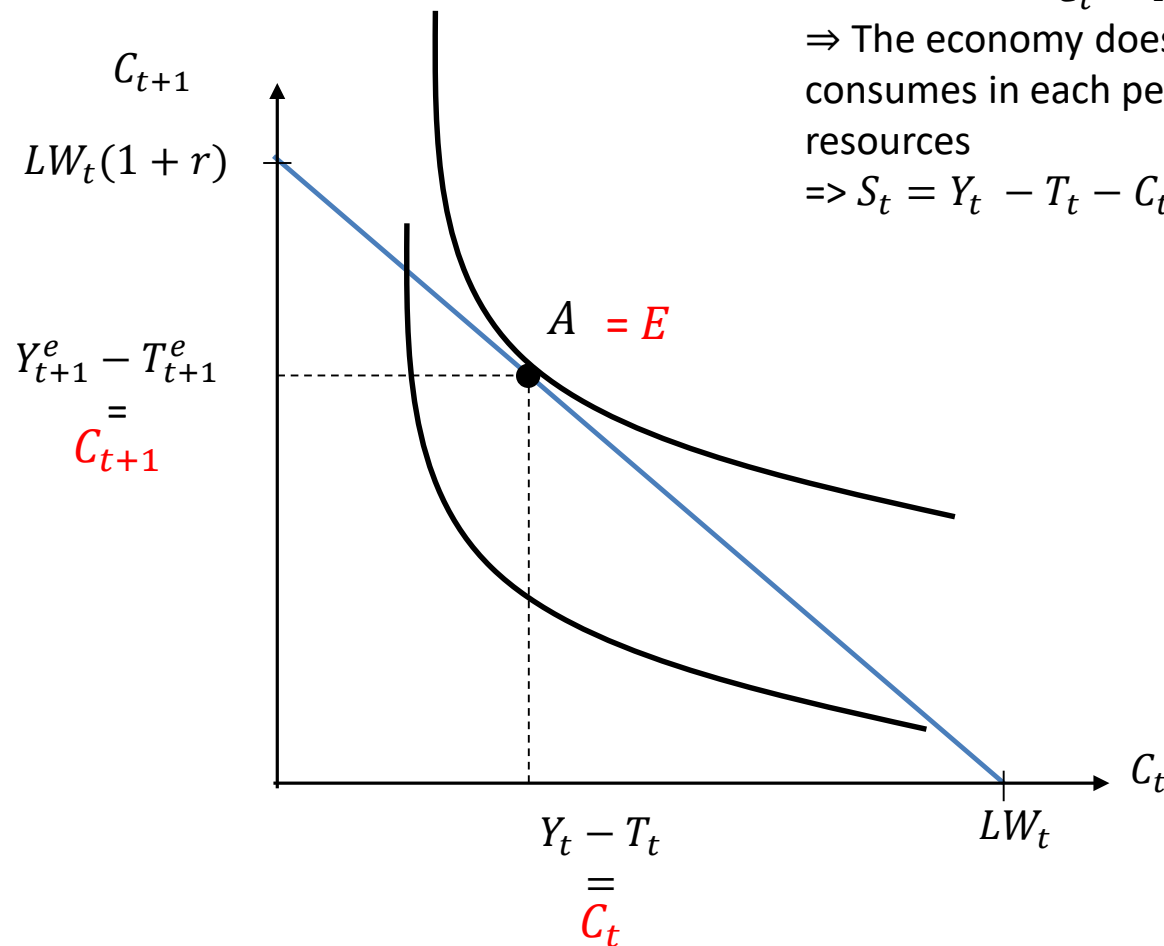
⇔ It is the point of tangency between the IBC and the highest attainable indifference curve

1) No saving

$$C_t = Y_t - T_t$$

⇒ The economy does not save and consumes in each period all the available resources

$$\Rightarrow S_t = Y_t - T_t - C_t = 0$$

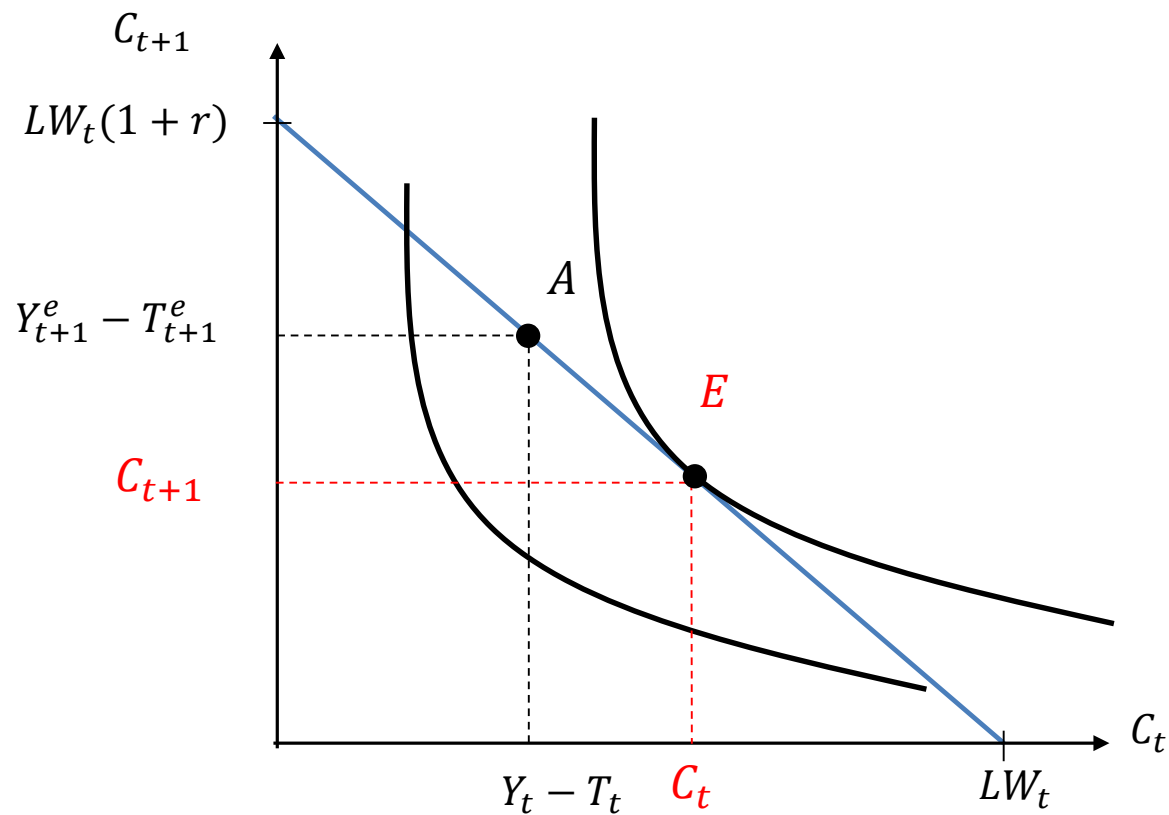


2) Borrower economy

$$C_t > Y_t - T_t$$

$\Rightarrow$ Negative saving

$$\Rightarrow S_t = Y_t - T_t - C_t < 0$$

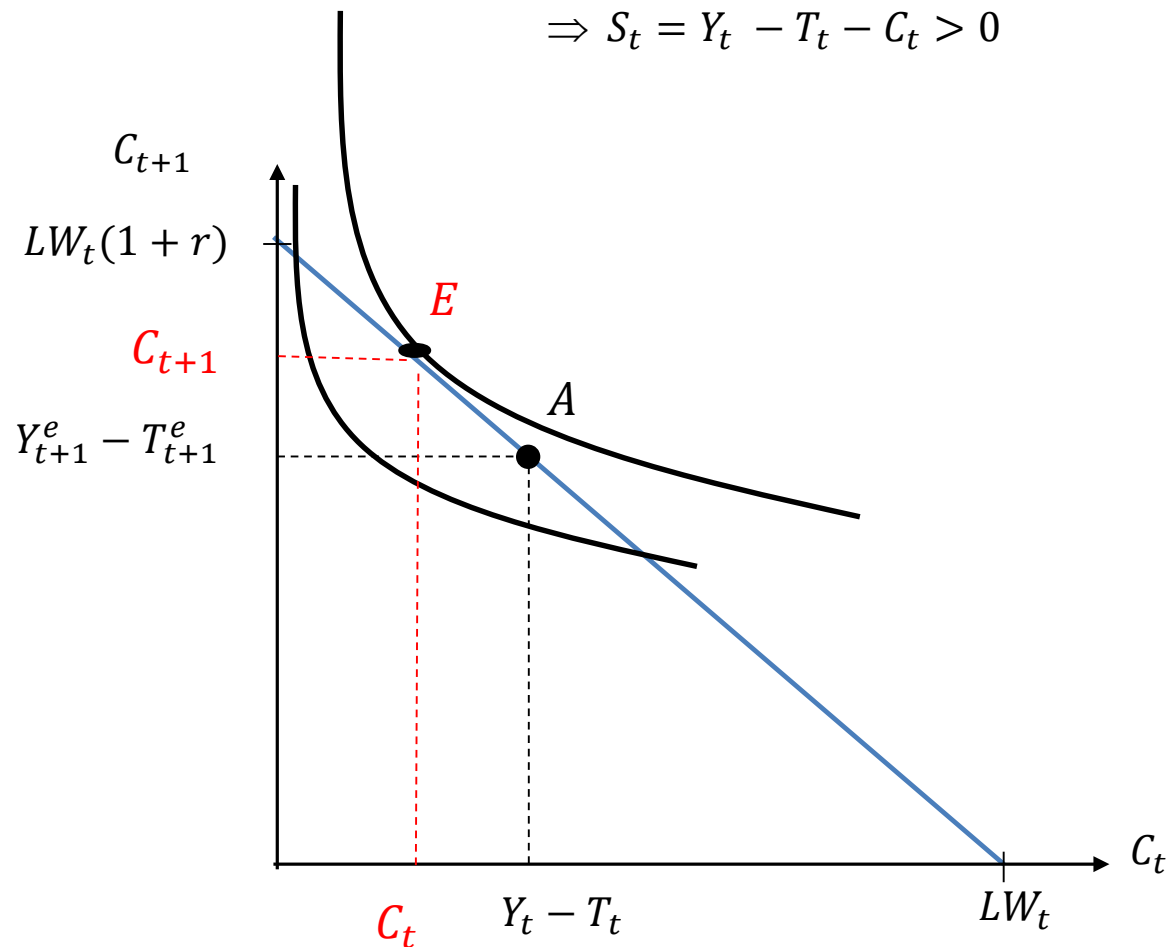


3) Saver economy

$$C_t < Y_t - T_t$$

$\Rightarrow$ Positive saving

$$\Rightarrow S_t = Y_t - T_t - C_t > 0$$



Does the economy save or borrow in the current period?

Being a saver or a borrower depends on:

- Preferences
- Disposable income today and tomorrow
- The interest rate
 - Opportunity cost of current consumption



When exogenous shocks hit the economy and alter these elements, the optimal choice changes



Since C represents about 70% of GDP, even small changes in aggregate C can have significant impacts on the real economy.
⇒ It is important to quantify these changes in order to design more effective policy interventions

COMPARATIVE STATICS

What happens when disposable income changes?

The effect on C depends on the intensity of the change in disposable income (how large it is and how long it lasts)



PERMANENT
VARIATIONS:
Larger change in C_t
 $\Leftrightarrow$ higher marginal
propensity to
consume

TEMPORARY
VARIATIONS:
smaller changes in C_t
 $\Leftrightarrow$ lower marginal
propensity to
consume

What happens when r changes?

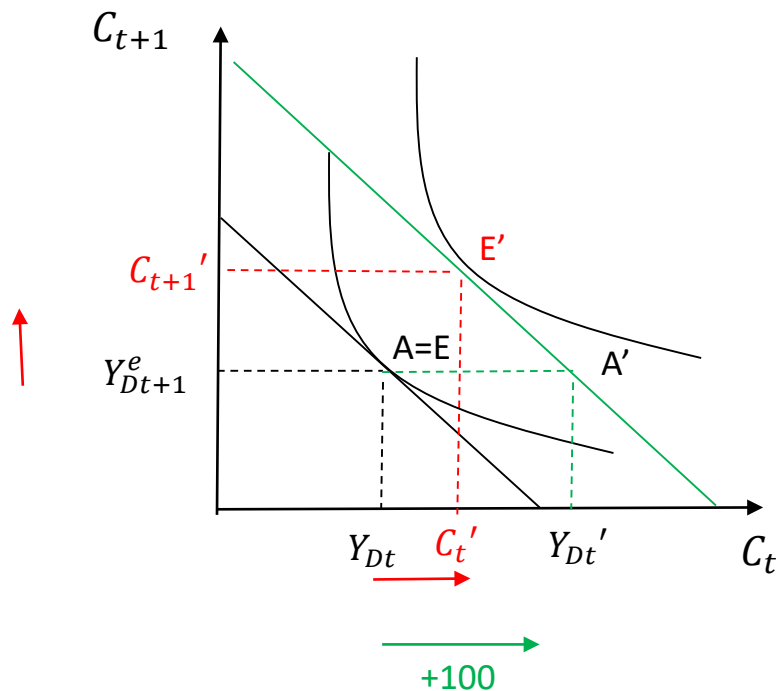
1) Temporary increase in disposable income

Consider an economy with no saving or borrowing, and assume the government reduces taxes by 100 only in the current period $\Leftrightarrow$ a temporary expansionary fiscal policy

What is Impact on consumption choices?

Disposable income increases only in the current period

- $\Delta(Y_t - T_t) > 0$
- $\Delta(Y_{t+1}^e - T_{t+1}^e) = 0$



But consumption opportunities $\uparrow$ in both periods $\Rightarrow$ $\uparrow$ permanent income (LW_t)

- $C_t \uparrow$ but $\Delta C_t < \Delta Y_{Dt} \Rightarrow S_t \uparrow$
- $C_{t+1} \uparrow$

The benefits of having more resources today are transmitted to the future:

- $\Rightarrow$ Current saving increases to enable higher consumption in the future
- $\Rightarrow$ Part of the current increase in income is saved to increase consumption tomorrow
- $\Rightarrow$ Forward-looking consumption smoothing

The size of the change in current and future consumption depends on the marginal propensity to consume

2) Increase in expected future disposable income

Consider an economy with no saving or borrowing, and assume the government announces a reduction in taxes by 100 for the next period $\Leftrightarrow$ expected expansionary fiscal policy

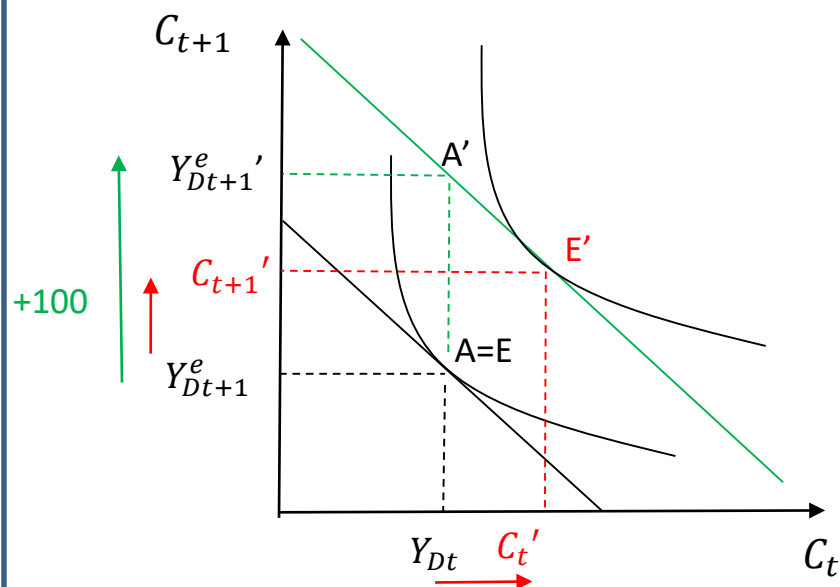
What is Impact on consumption choices?

Disposable income increases only in the next period

- $\Delta(Y_t - T_t) = 0$
- $\Delta(Y_{t+1}^e - T_{t+1}^e) > 0$

But consumption opportunities $\uparrow$ in both periods $\Rightarrow$
 $\uparrow$ permanent income (LW_t)

- $C_t \uparrow$ and $S_t \downarrow$ since $\Delta C_t > \Delta Y_{Dt}$
- $C_{t+1} \uparrow$ but $\Delta C_{t+1} < \Delta Y_{Dt+1}^e$



The benefits of having more resources tomorrow are transmitted to the present:

- $\Rightarrow$ Even if today actual resources are unchanged, we can consume more by borrowing/dissaving
- $\Rightarrow$ Current saving decreases to allow higher consumption today
- $\Rightarrow$ Part of the expected future income increase is used to finance higher consumption now
- $\Rightarrow$ Consume more today anticipating higher expected future disposable income
- $\Rightarrow$ Backward-looking consumption smoothing

The size of the change in current and future consumption depends on the marginal propensity to consume

3) Permanent increase in disposable income

Consider an economy with no saving or borrowing, and assume the government announces a reduction in current and future taxes by 100 $\Leftrightarrow$ expansionary fiscal policy in both periods

What is Impact on consumption choices?

Disposable income increases in the both periods

- $\Delta(Y_t - T_t) > 0$
- $\Delta(Y_{t+1}^e - T_{t+1}^e) > 0$

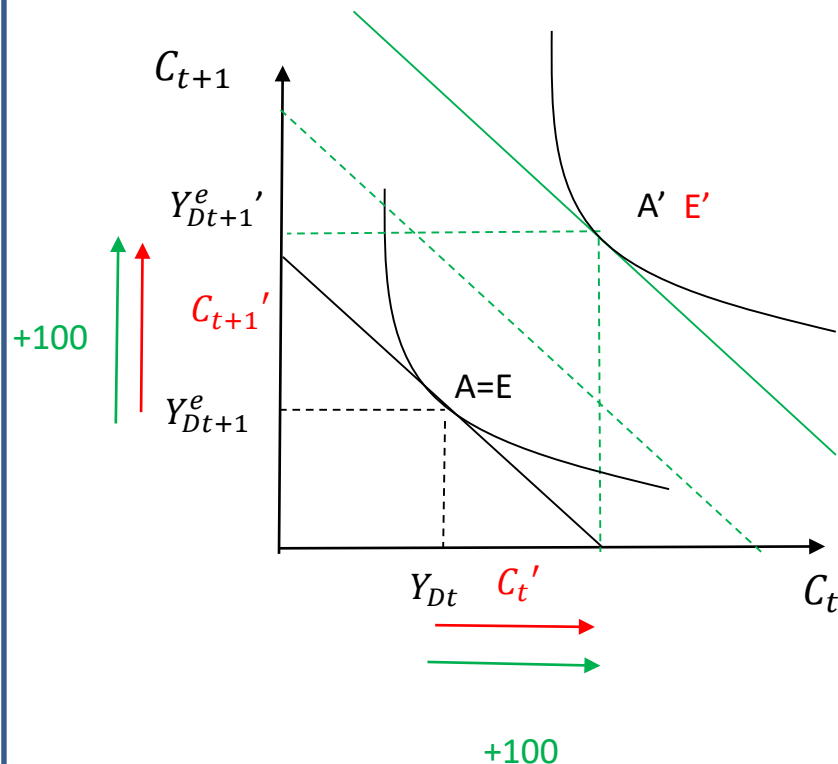
Consumption opportunities $\uparrow\uparrow$ in both periods $\Rightarrow$
 $\uparrow\uparrow$ permanent income (LW_t)

- $C_t \uparrow\uparrow$ and $S_t =$ since $\Delta C_t = \Delta Y_{Dt}$
- $C_{t+1} \uparrow\uparrow$ and $\Delta C_{t+1} = \Delta Y_{Dt+1}^e$

In principle, there is no change in saving in either period $\Leftrightarrow$ consumers simply consume the additional resources they have

- $S_t =$ since $\Delta C_t = \Delta Y_{Dt}$
- $S_{t+1} =$ and $\Delta C_{t+1} = \Delta Y_{Dt+1}^e$

Since the income increase is permanent, the benefit is evenly distributed across periods
 $\Leftrightarrow$ consumption rises in proportion to the total income change



4) Increase in interest rate

Assume the CB implements a contractionary MP

What happens if r increases?

- Several channels that act in different directions
 - Opportunity cost of consumption
 - Substitution effect
 - Income effect
 - Impact on the PDV of future expected income



We prefer to neglect these
microeconomic effects

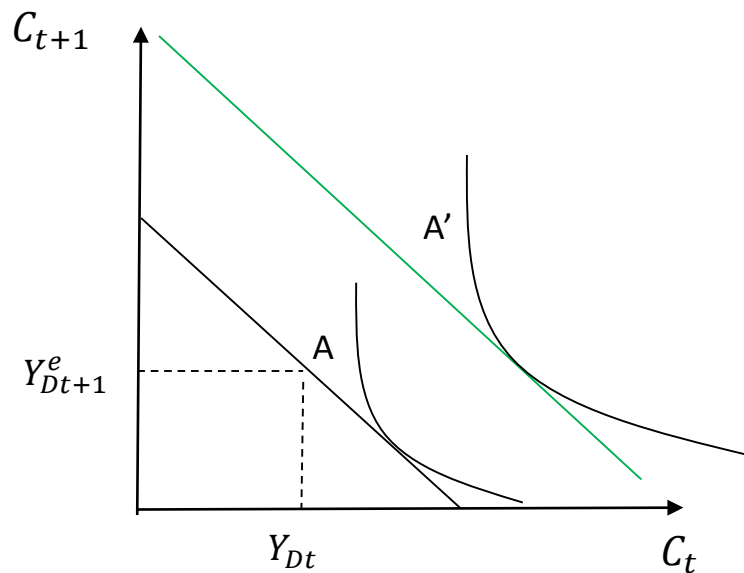
- In general, we can capture the impact of on the economy through:
 - Financial and housing wealth
 - Investment

OBSERVATIONS

1) What happens if agents hold also a financial and housing wealth?

$$C = f(W_t^T) = f(LW_t + FW_t + HW_t)$$

$$\text{IBC: } C_t + \frac{C_{t+1}}{1+r} = LW_t + FW_t + HW_t$$



↑ consumption possibilities since more resources are available =>
Possible to consume more in both periods

During the financial crisis both FW_t and $HW_t \downarrow \downarrow \Rightarrow C \downarrow \downarrow$ and $Y \downarrow \downarrow$

=> The effects were stronger in countries where a larger share of total wealth was held in non-human wealth

2) From the empirical evidence we have

- Proofs supporting the theory:

$$\Delta C_t < \Delta Y_{Dt}$$

$$\Delta C_t \neq 0 \text{ also if } \Delta Y_{Dt} = 0$$

- Proofs against the theory:

C_t changes more than predicted when Y_{Dt} changes

C_t changes less than predicted when Y_{Dt}^e changes

Why is it the case?

- Liquidity constraints: difficult/impossible to access credit markets Uncertainty
- Impatience

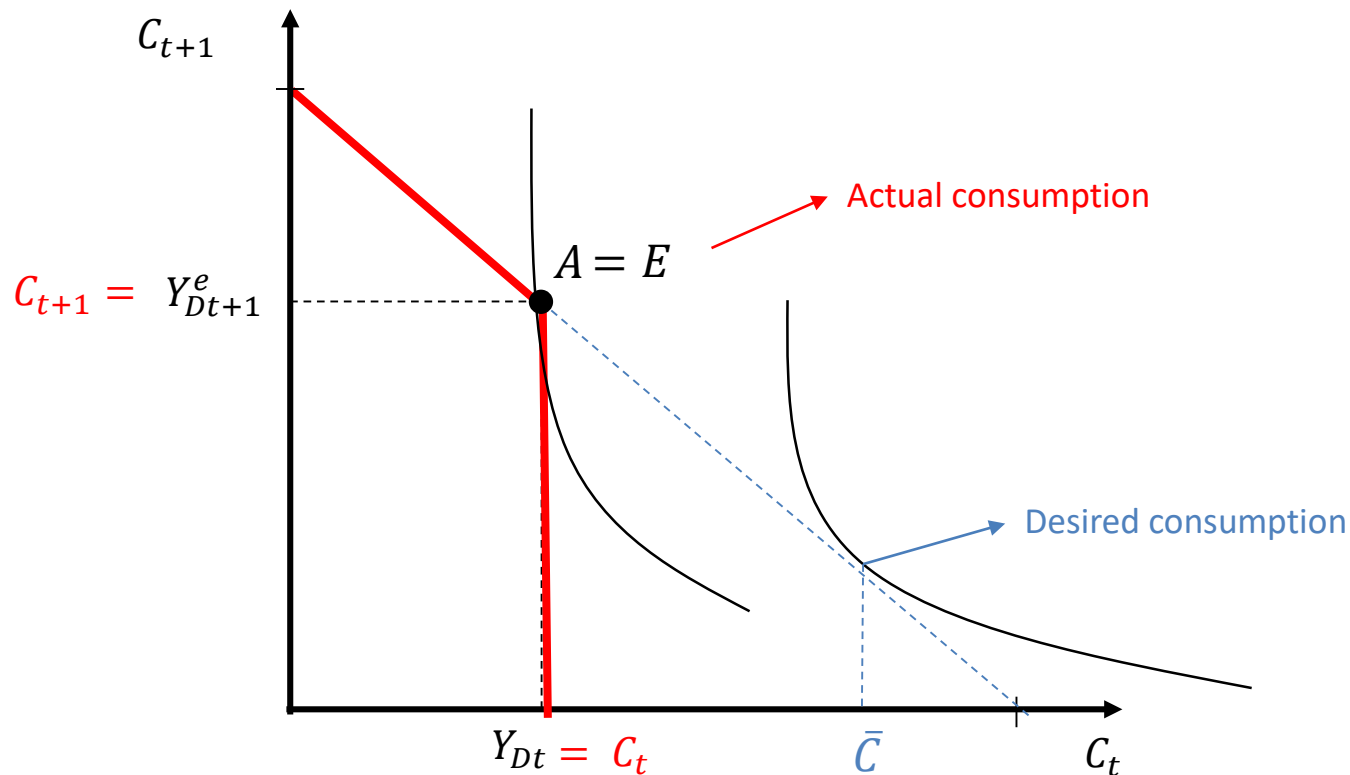
CREDIT CONSTRAINTS

Two types of constraints

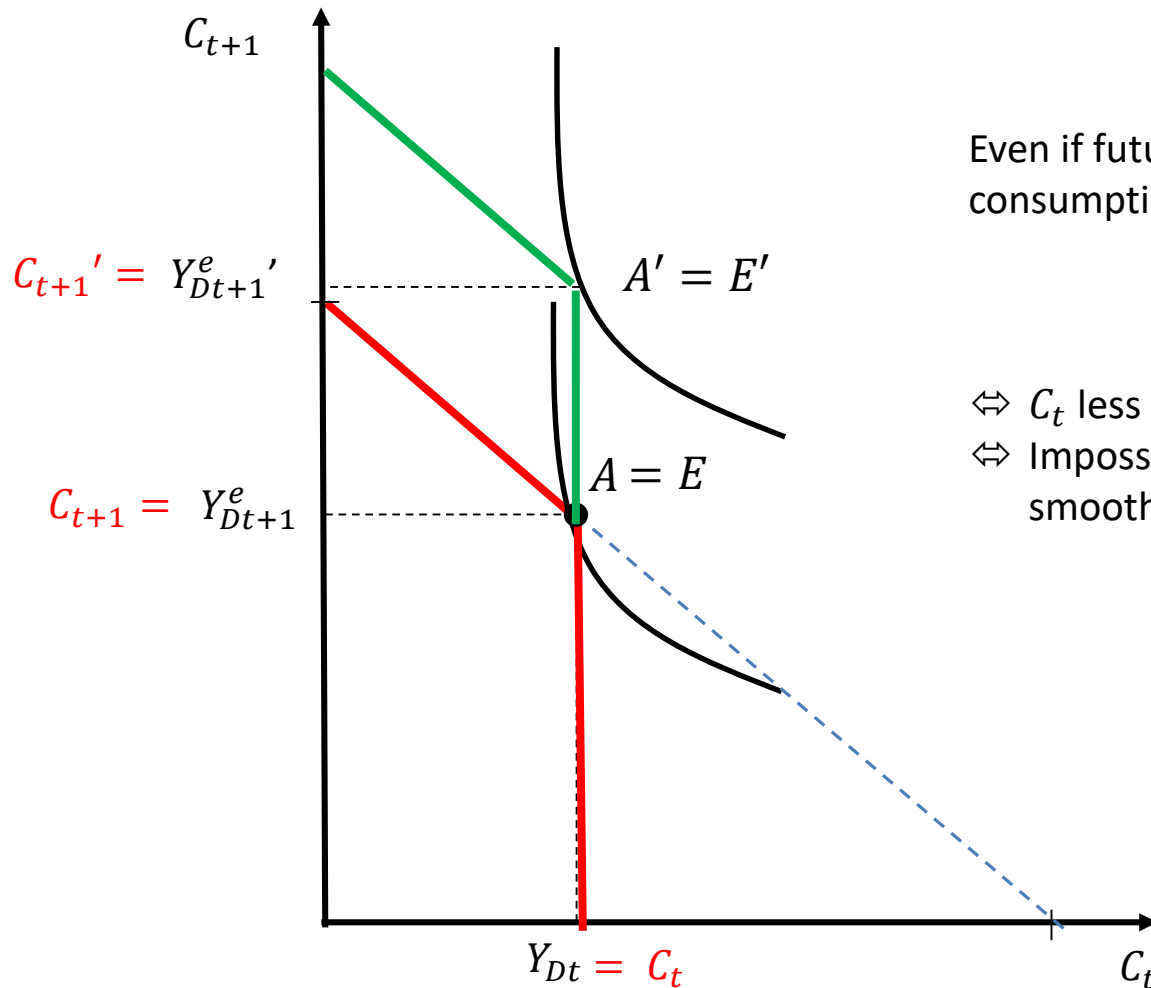
- Quantity constraints:
 - Total: borrowing is not allowed at all
 - Partial: there is an upper limit on how much can be borrowed
- Price constraints: it is possible to borrow but the borrowing rate is higher of the saving rate

1) Total quantity constraint

A credit-constrained individual would like to consume more than their endowment, but cannot borrow $\Rightarrow$ the suboptimal choice is to consume only their endowment



What happens if we expect an increase in future income?



Even if future income increases, current consumption does not change

$$\Delta C_t = 0$$

- $\Leftrightarrow C_t$ less sensitive to future income
- $\Leftrightarrow$ Impossible to make consumption smoothing backward

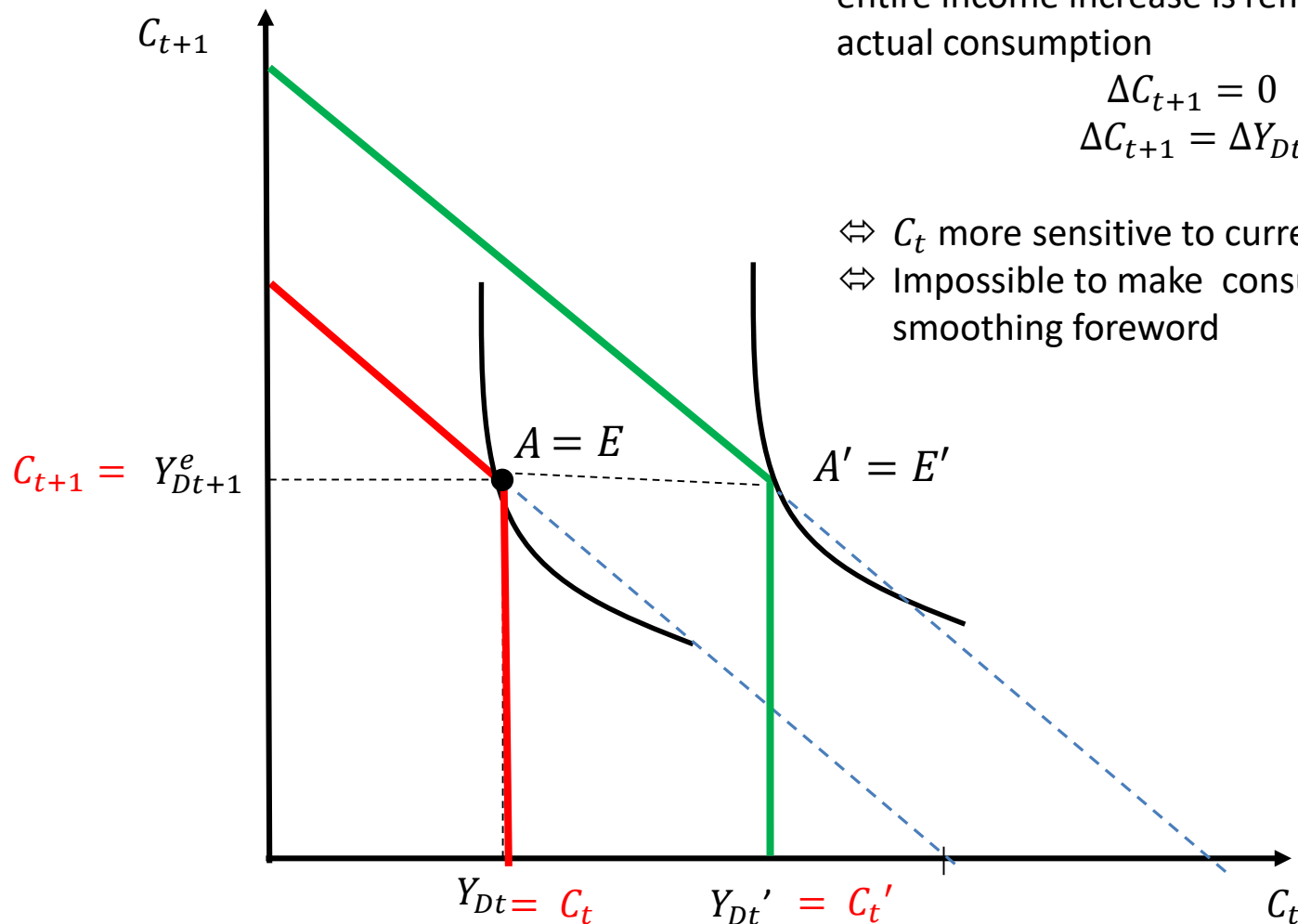
What happens if current income increases?

Even if permanent income increases, future consumption does not change and the entire income increase is reflected into actual consumption

$$\Delta C_{t+1} = 0$$

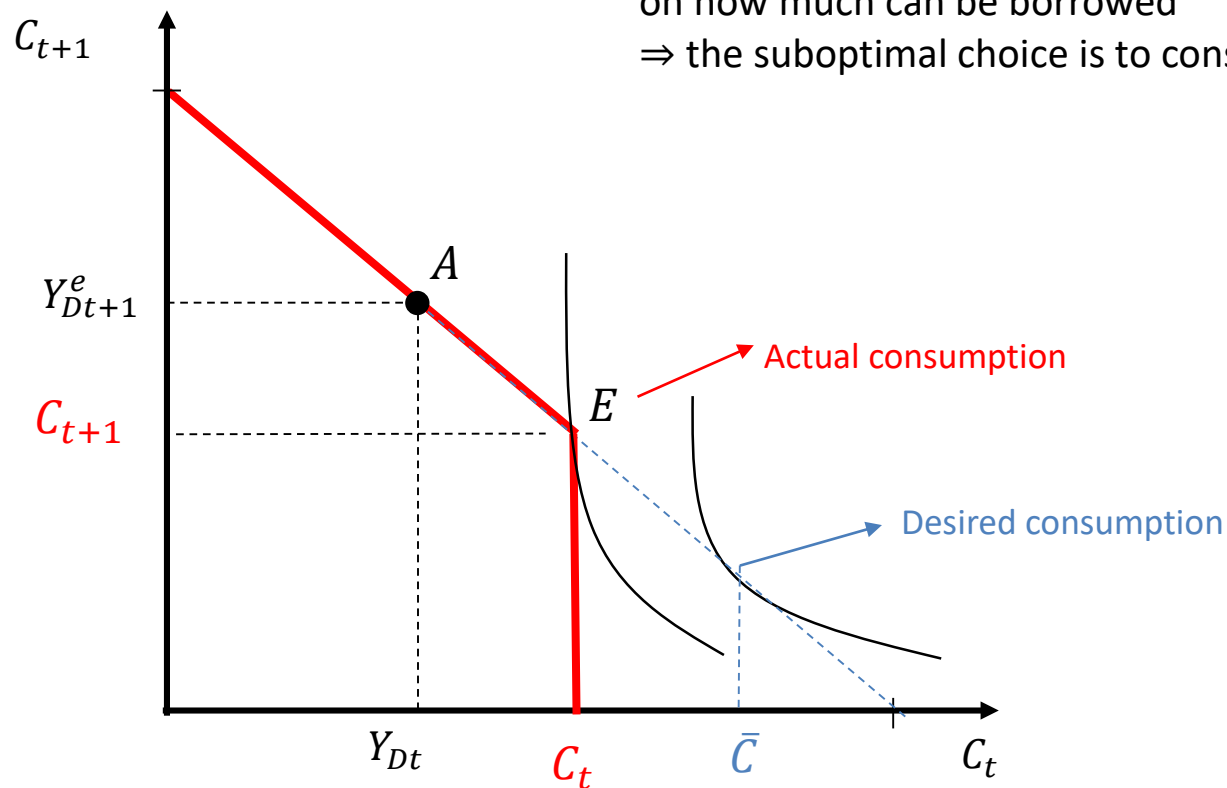
$$\Delta C_t = \Delta Y_{Dt}$$

- $\Leftrightarrow C_t$ more sensitive to current income
- $\Leftrightarrow$ Impossible to make consumption smoothing foreword



2) Partial quantity constraint

A credit-constrained individual would like to consume more than their endowment, but there is an upper limit on how much can be borrowed
⇒ the suboptimal choice is to consume less

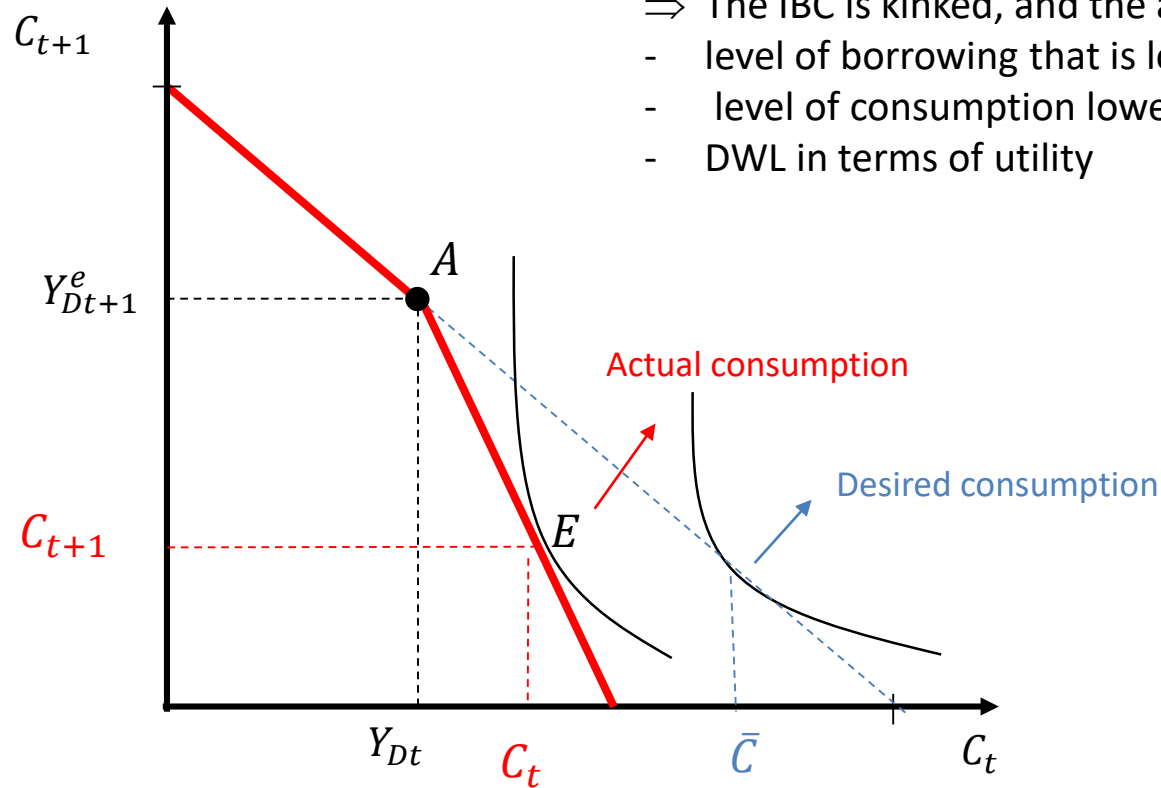


Same results as before: MPC to current income is higher and MPC to future income is lower

3) Price constraint

A price-constrained individual faces a borrowing rate that is higher than the saving rate: $r+x>r$

- ⇒ The IBC is kinked, and the actual choice implies
- level of borrowing that is lower than desired
 - level of consumption lower than the desired
 - DWL in terms of utility



Same results as before

SUMMING UP

- Foresighted consumer
- Consumption smoothing
- Higher impact of current disposable income

$$C_t = C(Y_t - T_t, Y_{t+1}^e - T_{t+1}^e, FW_t, HW_t, r_t)$$

- In some cases possible to have also

$$C_t = C(Y_t - T_t, Y_{t+1}^e - T_{t+1}^e)$$

QUICK CHECKS

TRUE OR FALSE

If the share of the population facing liquidity constraints is higher in Italy than in France, aggregate consumption will respond more to increases in current disposable income and less to increases in expected future income.

TRUE

Credit Constraints

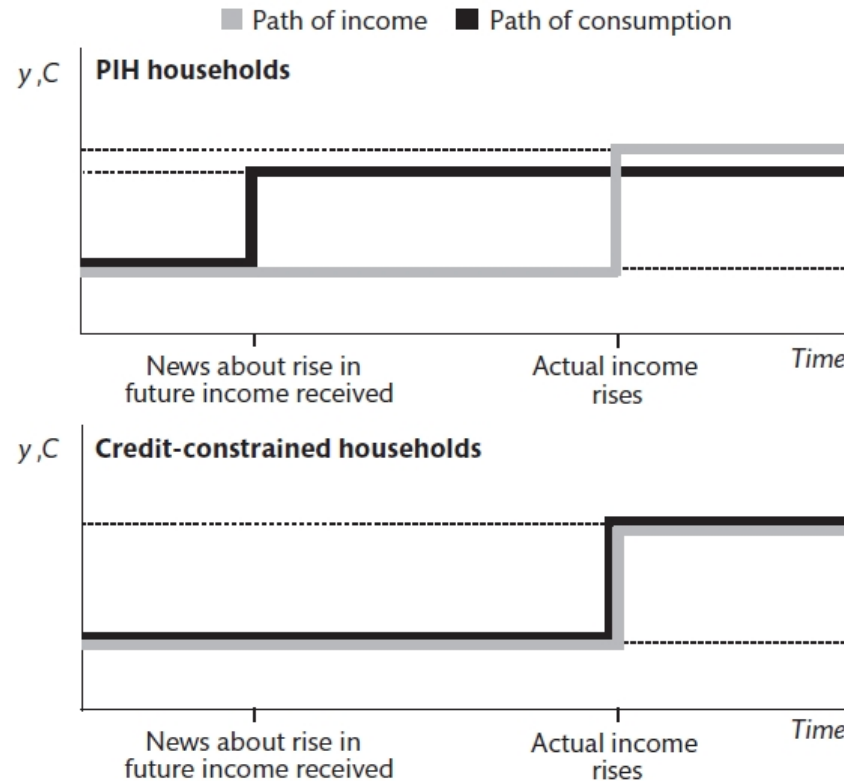


Figure 1.8 The consumption response of PIH and credit-constrained households to an anticipated rise in income.

Excess sensitivity: impatience

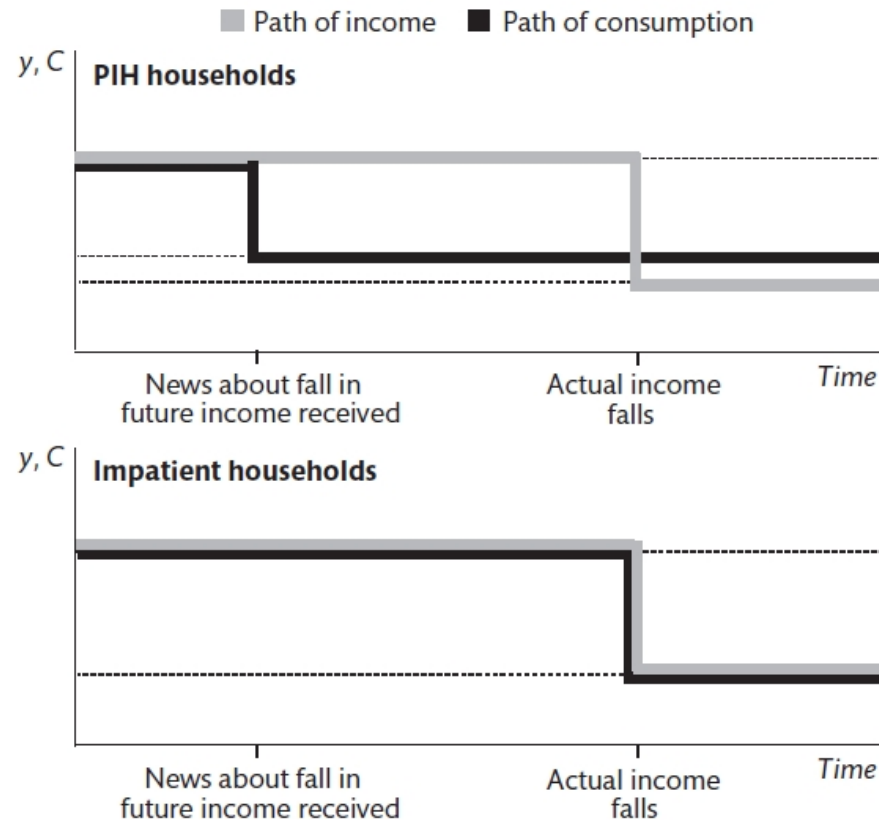


Figure 1.9 The consumption response of PIH and impatient households to an anticipated fall in income.

QUICK CHECKS

According to the intertemporal consumption model, desired consumption is:

- a) Equal to current income
- b) Equal to future expected income
- c) Smooth over time based on total lifetime wealth**
- d) Maximized when all income is consumed in the present

QUICK CHECKS

When individuals face total quantity liquidity constraints, their marginal propensity to consume:

- a) increases for future income
- b) decreases for current income
- c) Is zero for future income**
- d) none of the previous answers

QUICK CHECKS

In a liquidity trap with zero inflation, what happens if the government increases current taxes and decreases future taxes by the same amount?

- a) Consumption decreases because current disposable income falls
- b) Consumption increases because future taxes are lower
- c) Consumption remains unchanged**
- d) Consumption increases because monetary policy for sure will offset the shock

QUICK CHECKS

Temporary increases in income lead to:

- a) Large increases in consumption
- b) No change in consumption
- c) Smaller increases in consumption compared to permanent changes**
- d) Identical effects as permanent increases

TAKE AWAY AND KEY CONCEPTS

- Total lifetime wealth
 - Human wealth
 - Financial wealth
 - Housing wealth
- Theoretical Approaches
 - Keynesian
 - Intertemporal (micro-founded)
- Intertemporal Consumption
 - optimal path
 - intertemporal budget
 - preferences
- Temporary income increase
- Expected future income increase
- Permanent income increase
- Marginal propensity to consume

INVESTMENT AND EXPECTATIONS

How is aggregate investment in physical capital determined?

Firms decide to invest in new machineries, plants, equipment, vehicles based on

- Profitability (sales and profits)
- Borrowing rate

Two approaches

1. **Traditional Keynesian theory**: current investment depends on current variables

$$I_t = f(Y_t, r_t + x_t)$$

2. **Intertemporal theory**: current investment depends on the expected stream of profits from the period of capital acquisition to the disposal

$$I_t = f(V(\Pi_t^e))$$



Physical capital generates profits over multiple periods, depending on its useful life and profitability declines over time due to depreciation from wear and tear and technological obsolescence.

Investment decision implies a cost benefit analysis:

firms invest only if it is economically convenient

=> Net profits making the investment should be higher than profits without making the investment

Benefit

PDV of expected profits

Cost

Cost of the investment

Assumptions:

- P_k = cost of 1 unit of capital
- The firm invests in period t but production starts in $t+1$
=> first profits next period
- expected profits (per unit) in each period Π^e
- Physical capital

- has an useful life
- over time loses value due to depreciation that starts at $t+2$

Constant depreciation rate δ =
rate at which capital good loses its
usefulness over time
(obsolescence)
 $\Leftrightarrow$ Over time physical capital is less
effective in generating profits

δ : 4 – 15 % for machinery
2 – 4 % for real estate and equipment

t

1 unit of physical capital is installed

$t + 1$

Profits are obtained

$t + 2$

Devaluation starts and value of capital is $(1 - \delta)$, generating profits

$t + 3$

Devaluation continues and value of capital is $(1 - \delta)^2$, generating profits



Profits flow

Π_{t+1}^e

$(1 - \delta)\Pi_{t+2}^e$

$(1 - \delta)^2\Pi_{t+3}^e$

....

Value in t

$$\frac{\Pi_{t+1}^e}{1 + r_t}$$

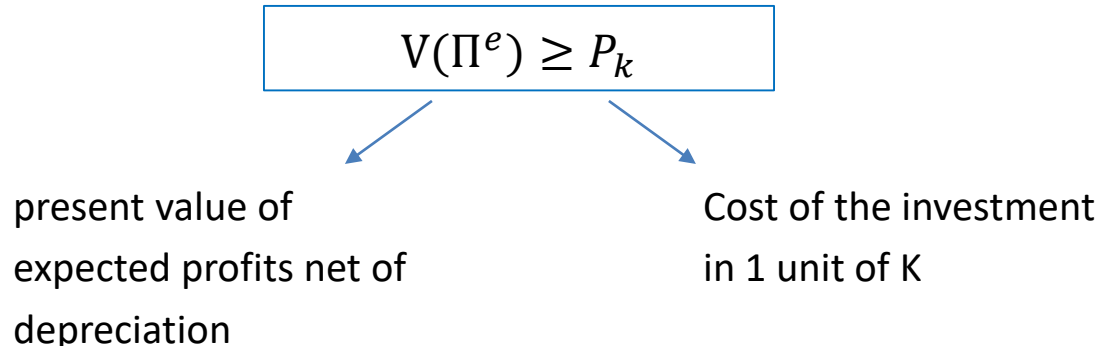
$$\frac{\Pi_{t+2}^e(1 - \delta)}{(1 + r_t)(1 + r_{t+1}^e)}$$

$$\frac{\Pi_{t+3}^e(1 - \delta)^2}{(1 + r_t)(1 + r_{t+1}^e)(1 + r_{t+2}^e)}$$

....

→ PDV of expected profits net of depreciation over the capital lifecycle
 $= V(\Pi^e)$

At time t investment is made if

$$V(\Pi^e) \geq P_k$$


```
graph TD; A["V(Π^e) ≥ P_k"] --> B["present value of expected profits net of depreciation"]; A --> C["Cost of the investment in 1 unit of K"]
```

present value of
expected profits net of
depreciation

Cost of the investment
in 1 unit of K

Or equivalently if

$$V(\Pi^e) - P_k \geq 0$$

Net profits for an intertemporal
perspective are not negative

When does investment increases?

When benefits increases w.r.to costs $\Rightarrow$ more convenient to invest

- If given the cost, we expect higher profits

$$\text{If } \Pi^e \uparrow \Rightarrow V(\Pi^e) \uparrow \Rightarrow I_t \uparrow$$

- If is less costly to finance the investment in the market

$$\text{If } r \downarrow, r^e \downarrow$$

$\Rightarrow$ the opportunity cost of investing $\downarrow$

$$\Rightarrow V(\Pi^e) \uparrow \Rightarrow \text{higher profitability of current investment} \Rightarrow I_t \uparrow$$

- If the depreciation is lower

$\text{If } \delta \downarrow \Rightarrow$ physical capital loses value slowly

$\Rightarrow$ given its cost the benefits from investing are higher $\Rightarrow V(\Pi^e) \uparrow$

$$\Rightarrow \text{more convenient to invest} \Rightarrow I_t \uparrow$$

- If physical capital cheaper

$\text{If } P_k \downarrow \Rightarrow$ given unitary profits, net profits $\uparrow$

$\Rightarrow$ benefits from investing $\uparrow$

$$\Rightarrow \text{more convenient to invest} \Rightarrow I_t \uparrow$$

The aggregate investment function could be written synthetically as

$$I_t = f(V(\Pi^e) , P_k) = f(\overset{+}{\Pi^e} , \overset{-}{r} , \overset{-}{r^e} , \overset{-}{\delta} , \overset{-}{P_k})$$

Special case of the investment function with static expectations (future=present)

$$\begin{aligned} \Pi_{t+1}^e &= \Pi_{t+2}^e = \Pi_{t+3}^e = \dots = \Pi \\ r_{t+1}^e &= r_{t+2}^e = r_{t+3}^e = \dots = r \end{aligned}$$

The investment function becomes

$$I_t = I (V(\Pi^e)) = I \left(\frac{\Pi}{r + \delta} \right)$$

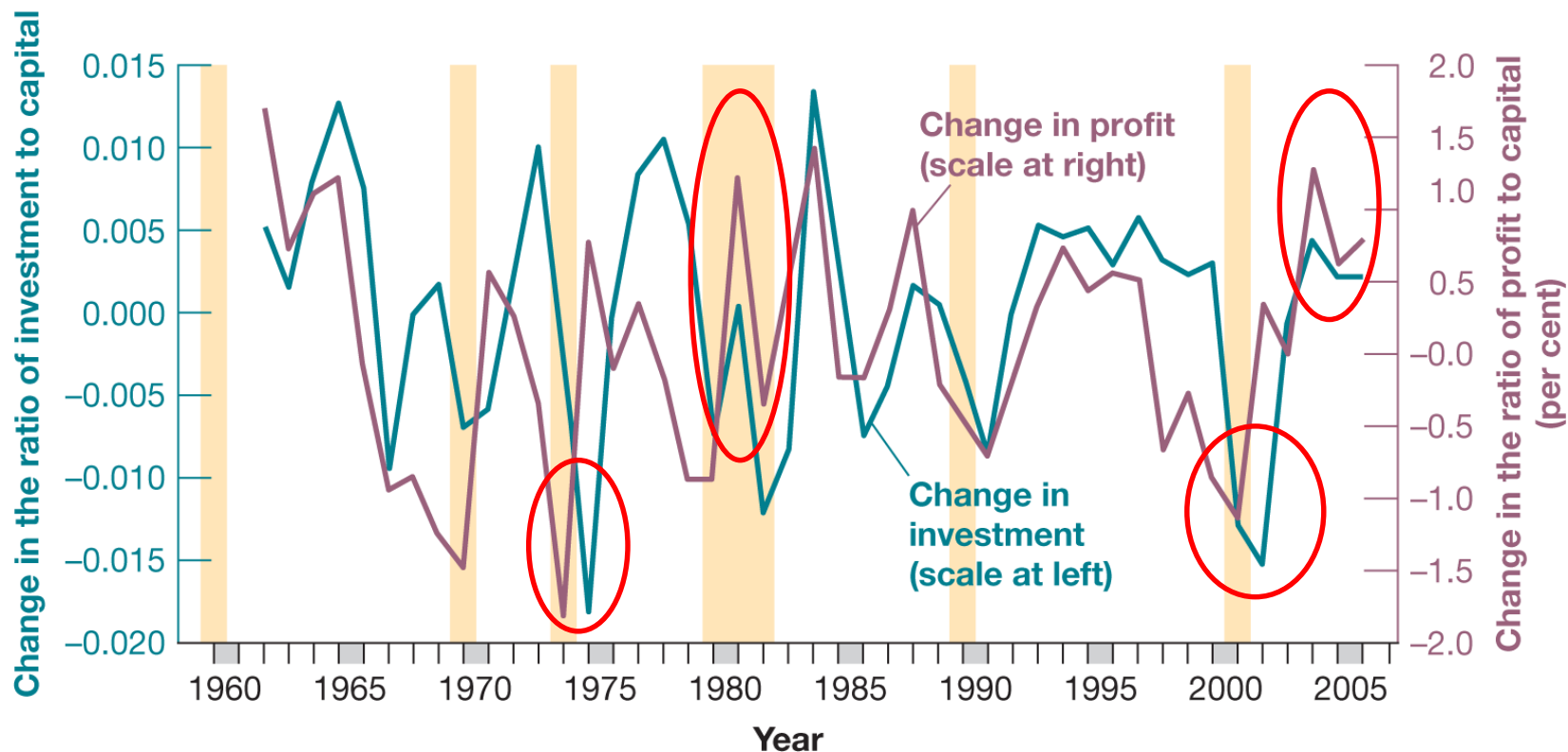
➔ Profit per unit of invested capital

➔ Rental (user) cost of capital in each period:
It includes the borrowing cost and the depreciation

Investment depends on the ratio of profit to the user cost:

The higher the profit, the higher the level of investment

The higher the user cost, the lower the level of investment



Capital stock measured by non-financial assets; profit is constructed from net operating surplus, taxes and transfers, Bureau of Economic Analysis.

Empirical evidence show that
investment and current profits move together

If $\Pi_t \uparrow \Rightarrow I_t \uparrow$ and vice versa

Why is investment very sensitive to current profits, more than to future ones as predicted by our model?

If a firm's **current profit** is low:

- it can get the funds it needs only by borrowing if it wants to buy new machines
- it might have difficulty borrowing even if it wants to invest

More realistic behavioral equation for investment

$$I_t = f(\Pi_t, V(\Pi^e), P_k)$$

+ + -

Current economic macro
dynamic

$$Y_t \uparrow \Rightarrow \Pi_t \uparrow \Rightarrow I_t \uparrow$$

During expansions, firms sell more, generate higher profits, and tend to invest more

Expected economic macro
dynamic

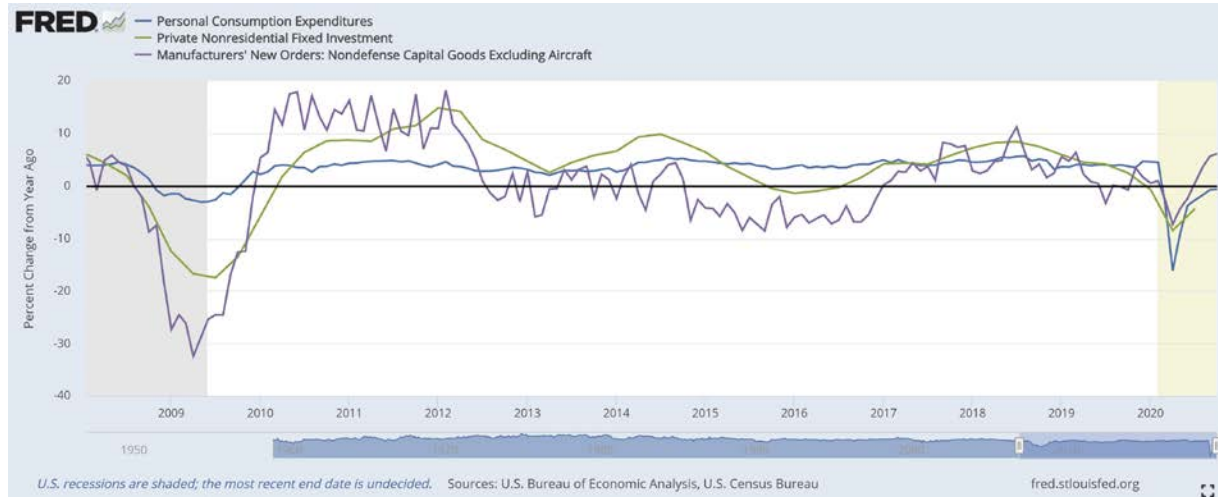
- $Y^e \uparrow \Rightarrow \Pi^e \uparrow \Rightarrow I_t \uparrow$
- $r \uparrow \Rightarrow I_t \downarrow$
- $r^e \uparrow \Rightarrow I_t \downarrow$
- $\delta \uparrow \Rightarrow I_t \downarrow$

When firms expect expansions, expect to sell more and to obtain higher profits, and have higher incentive to invest today

When they expect contractionary MP, PDV of expected profits increases and current investment is more attractive

Empirical evidence

- Investment reacts more to permanent changes in sales (GDP) than to temporary changes
 - meaning the marginal propensity to invest is higher with permanent shocks.
- Co movements of consumption and investment over the economic cycle
- Investment (I) is more volatile than consumption (C)



QUICK CHECKS

If the government announces a permanent tax cut starting next year and the CB has a target in terms on money supply, what happens to investment today?

- a) It decreases because future profits are less relevant for current decisions
- b) It increases because expectations of higher future profits increases PDV of profits
- c) It could remain unchanged**
- c) It remains unchanged because investment depends only on current profits

QUICK CHECKS

If the depreciation rate of physical capital increases, what happens to investment?

- a) It increases because machines need to be replaced more often
- b) It remains unchanged because it depends only on future profits
- c) It increases due to higher fiscal incentives
- d) It decreases because the user cost of capital rises**

QUICK CHECKS

If firms expect a permanent decrease in aggregate demand, how do investments react today?

- a) **They decrease because expected future profits fall**
- b) They increase to compensate for lower expected sales
- c) They remain unchanged because investment depends only on current conditions
- d) They increase due to lower expected inflation